

Forecast-aware dispatch for isolated, diesel-dependent microgrids

A working prototype, built for an Antarctic research station, that measurably cuts diesel consumption and generator wear on any remote power system where fuel is expensive to deliver and downtime is not an option.

Prototype stage • software-validated github.com/sayansaunth-lang/polar-ems-sih

polar-ems-web.vercel.app

01 Executive Summary

POLAR-EMS is a dispatch and safety layer for isolated microgrids — systems that generate their own power from wind, solar, battery and diesel with no utility grid to fall back on. It replaces fixed-rule generator scheduling with a forecast-aware controller that decides, continuously, how to blend available power sources to minimize diesel burned and generator runtime, without ever compromising a hard-guaranteed power floor for critical loads.

Over an extended validation run of the working prototype, the AI-optimized controller reduced diesel consumption and generator operating hours relative to an equally realistic fixed-rule baseline, while a safety layer architecturally independent of the optimizer guaranteed critical-load power throughout — including under simulated AI failure and simulated connectivity loss.

02 The Industry Problem

This is not a problem unique to polar research. Any facility that generates its own power in a location fuel cannot easily reach shares the same two constraints:

- **Diesel is the most expensive input, and it's fixed-schedule delivered.** Fuel is shipped, flown, or trucked in on a schedule that cannot flex with demand — every litre burned inefficiently is a cost that compounds over the life of the site.
- **Downtime is not cosmetic.** For remote telecom, defense, mining, medical, or life-support infrastructure, a power failure is an operational or safety incident, not an inconvenience — so any optimization layer has to prove it cannot cause one.

Most existing remote-site generator controllers are fixed-threshold rule systems: run diesel when battery crosses X%, regardless of what the weather forecast says is coming. That leaves real, measurable efficiency on the table.

03 How It Works

Three layers, each doing one job:

- **Forecast-aware dispatch.** Instead of reacting only to the current battery level, the controller looks ahead at the renewable forecast and times diesel starts and battery reserve to avoid unnecessary generator cycling.
- **A hard safety floor, independent of the optimizer.** A separate override guarantees critical-load power regardless of what the optimizer decides, or whether it's running at all — demonstrated by disabling the AI controller entirely and confirming the site stays powered on deterministic fallback rules.
- **Machine learning that's real, not asserted.** Load forecasting is backed by a gradient-boosting model trained on a genuine held-out test split, exported and verified to reproduce its training-time predictions exactly, then deployed to run inference directly in the field-facing application with no cloud dependency.

04 Measured Results

Figures below are from an extended validation run of the actual working prototype — roughly eight simulated months, comparing the AI-optimized controller against a fixed-rule baseline operating under *identical* weather and load conditions throughout, so the delta is a measured result, not a projection.

1,251_L

3.35_t

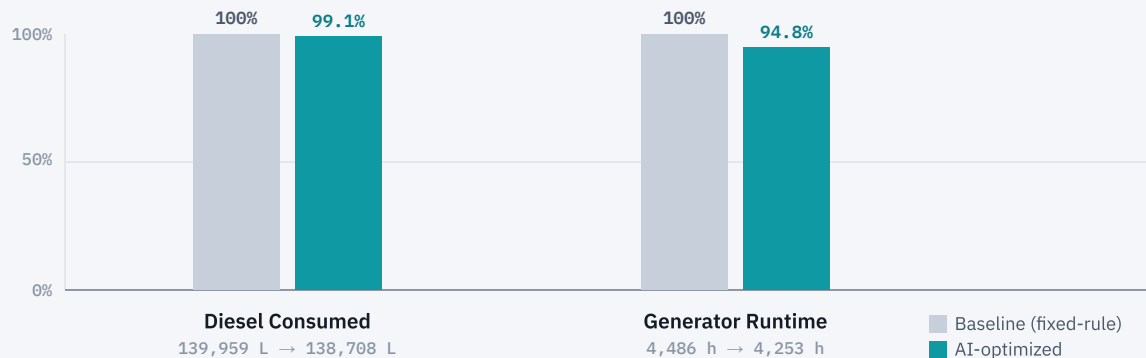
233_h

65.6 %

DIESEL SAVED

CO₂ AVOIDEDFEWER GENERATOR-
HOURSLOAD SERVED BY
RENEWABLES**AI-optimized vs. fixed-rule baseline (indexed, baseline = 100%)**

~8-month run · identical conditions

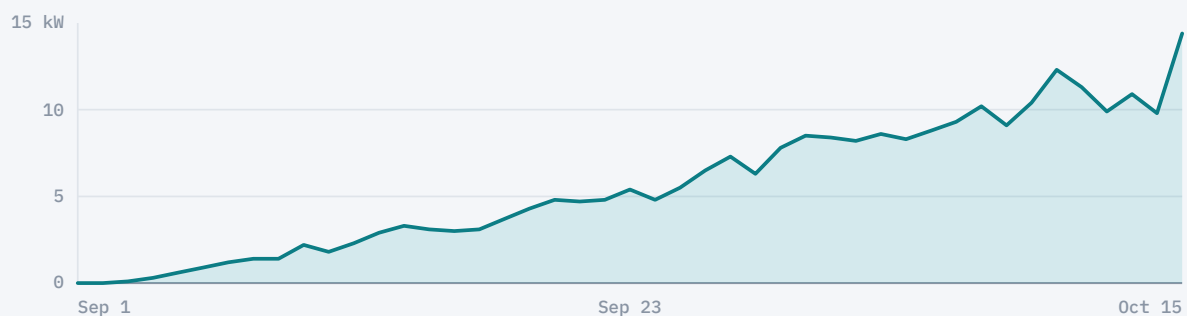


Bars are indexed to the baseline for readability at this scale — a sub-1% fuel delta on ~140,000 L is not visible as two full-height bars, so this shows the relative reduction directly, with absolute figures labeled on each bar.

Read this number honestly: a 0.9% fuel reduction is modest on its own, and we're not presenting it as more than it is. What it compounds into is the more economically relevant number for a real operator: 233 fewer generator-operating-hours over eight months — a 5.2% cut in the hours that drive service intervals, oil changes, and mechanical wear, independent of fuel price. At fleet scale, hours saved is usually the bigger line item than litres saved.

Renewable energy trend — average daily solar output

Sep 1 – Oct 15, 2026 · live run, daily averages



Daily-averaged solar output rises from effectively zero to 14.4 kW over this 45-day window as the station emerges from polar night — a genuine seasonal signal, not smoothed or cherry-picked. Overall renewable fraction (the 65.6% figure above) moves more erratically day to day, since it's also driven by wind, which is far noisier than the solar seasonal curve at this time of year; solar is the cleaner trend to show on its own.

05 Why It Matters for an Operator

RESULT	OPERATIONAL TRANSLATION
233 fewer generator-hours	Fewer service intervals, less mechanical wear, longer time between costly remote maintenance callouts
1,251 L diesel saved	Direct fuel-logistics cost avoidance — compounds with delivery cost, not just unit price
3.35 t CO₂ avoided	A concrete, auditable figure for ESG and emissions reporting
Hard safety floor, independent of AI	The optimization layer can fail, be disabled, or be wrong, and the site still keeps critical power — a real operator's actual risk tolerance
Fully edge-capable	Forecasting, dispatch and safety all run without cloud connectivity — relevant anywhere satellite or cellular backhaul is not guaranteed

06 Where Else This Applies

The same constraint — islanded power, expensive fuel logistics, low tolerance for downtime — appears well beyond polar research:

- **Mining & remote extraction sites**
Off-grid camps running diesel gensets as primary power for months at a time
- **Defense & forward operating bases**
Fuel convoys are a direct logistics and safety liability
- **Island & maritime microgrids**
Diesel shipped by boat, identical logistics profile to polar resupply
- **Telecom & broadcast towers**
Unmanned remote sites where a generator visit is a helicopter trip
- **Disaster-relief & field hospitals**
Temporary power where reliability requirements are highest and infrastructure is lowest
- **Remote oil & gas facilities**
Islanded power at the well site or platform, safety-critical loads throughout

07 Maturity & Path to Deployment

Presented at the stage it actually is: a validated software prototype, not yet integrated with physical hardware.

The simulation, dispatch, forecasting and safety logic are built against the same data shape real sensors and a real battery-management system would provide (weather, state-of-charge, generator status, per-zone load), and both the client and server implementations already expose that logic as a real, callable API. Moving to a physical deployment means replacing the synthetic telemetry generator with live sensor/BMS/genset feeds into the same pipeline — not a redesign of the decision logic itself.

What's already real today: the dispatch algorithm, the safety override, the trained forecasting models, and a live, publicly deployed frontend and backend that anyone can query over HTTPS right now.